Units

1 State whether the following units can be used to measure length:

a Liters, L

b Centimeters, cm

c Cubic centimeters, cm^3

d Kilometers, km

e Square meters, m^2

f Millimeters, mm

2 State whether the following units can be used to measure area:

a Centimeters, cm

b Cubic kilometers, km^3

c Hectares, ha

d Liters, L

e Square meters, m^2

f Millimeters, mm

3 State whether the following units can be used to measure volume:

a Liters, L

b Cubic meters, m^3

c Cubic centimeters, cm^3

d Cubic kilometers, km^2

e Square meters, m^2

f Pounds, lbs

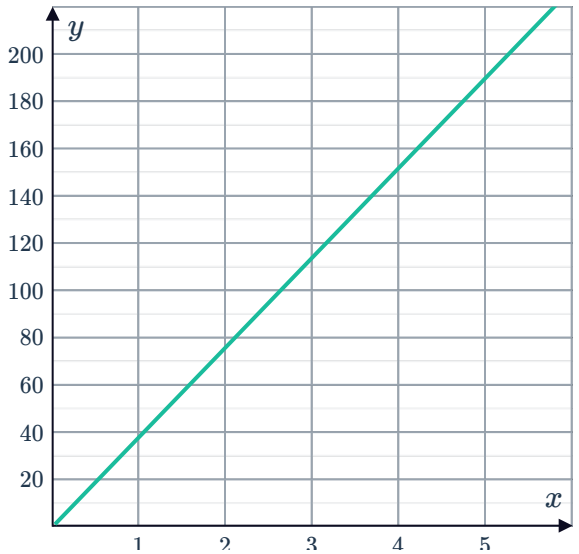
4 The manager of an electronics repair store wants to determine how much time, on average, his employees spend replacing one broken cellphone screen.
State an appropriate rate with which to calculate an answer to his question.

Units from a graph

5 The slope of the following graph has a unit of kilometers per hour:

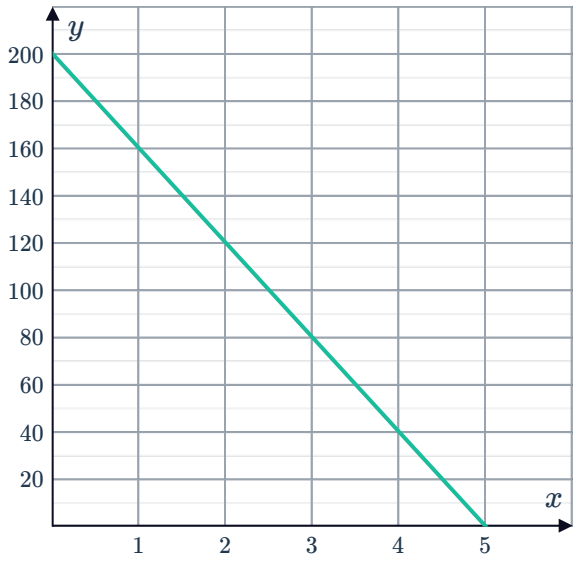
a State the unit on the x-axis.

b State the unit on the y-axis.



© ClipSAT: By Mr. Mohamed Abdallah +966597688647

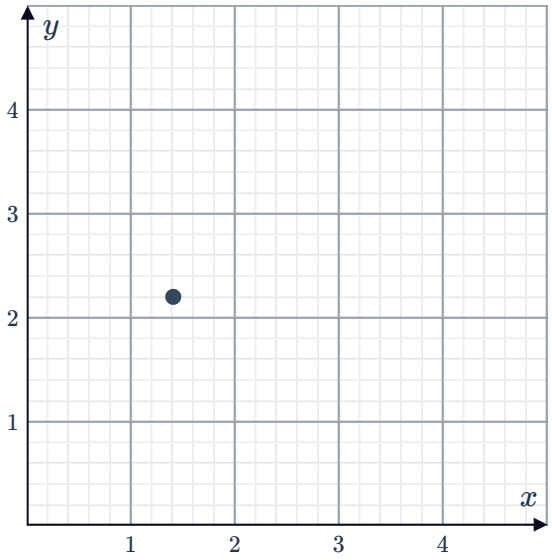
6 The slope of the following graph has a unit of yards per second:



a State the unit on the x-axis.

b State the unit on the y-axis.

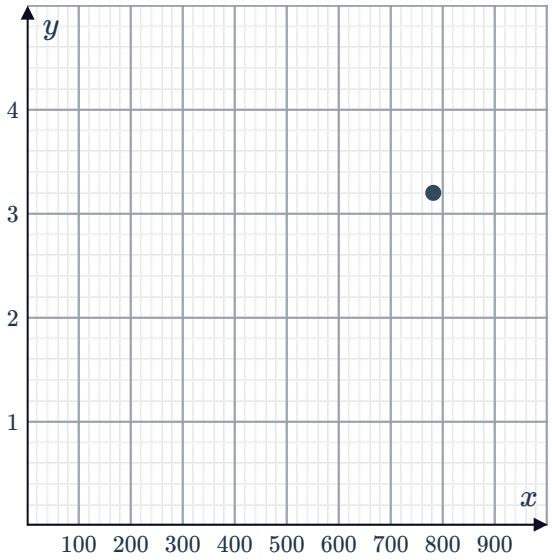
7 Adam plotted a point to represent material purchases, x, and the costs involved, y. When Adam bought 140 cm of material, it cost $\$2.20$.



a State the unit on the x-axis.

b State the unit on the y-axis.

8 Noah plotted a point to represent the amount of ingredients purchased, x, and the costs involved, y. When Noah bought 0.78 kg of ingredients, it cost $\$3.20$.

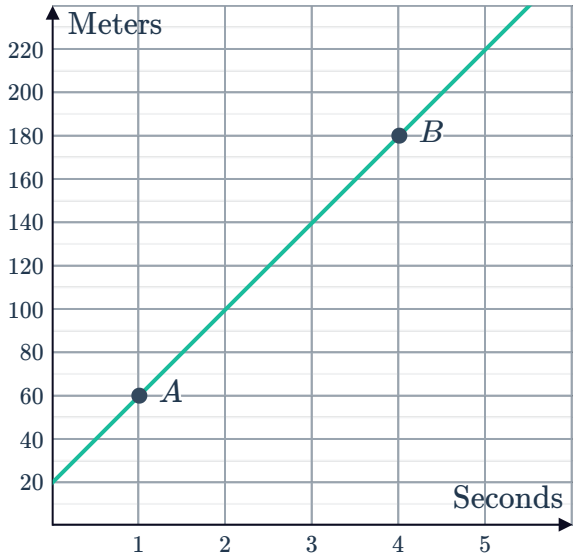


a State the unit on the x-axis.

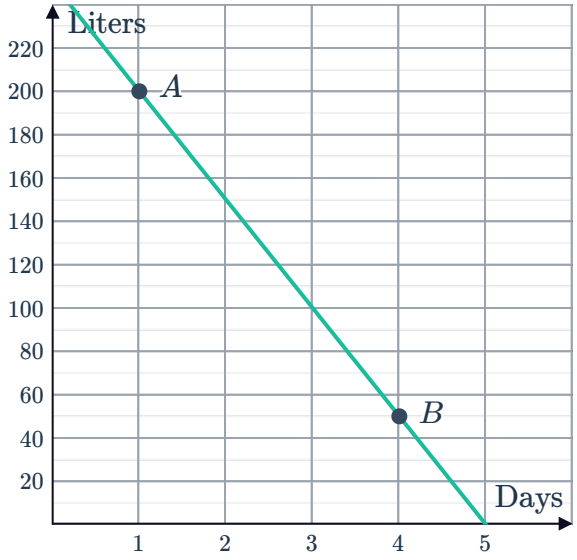
b State the unit on the y-axis.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

9 The slope between points A and B on the following graph was calculated and found to be 40.
What would be the correct unit for this measure?



10 The slope between points A and B on the following graph was calculated and found to be -50.
What would be the correct unit for this measure?



Units from formulas

11 The area of a rectangle is given by $A = l \times w$, where l is the length and w is the width.
What will the unit for area be if the length and width are in inches?

12 The speed, s, of an object is calculated using the formula $s = \frac{d}{t}$, where d is the distance traveled and t is the time taken.
Glen is carrying out a science experiment and measures the distance in millimeters and the time in hours. What will the unit for speed be?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

13 The distance an object travels can be calculated using the formula $d = s \times t$, where s is the speed of the object and t is the time taken to travel that distance.
If Tracy records the speed in feet per hour, what unit of distance will her calculation have?

14 The volume of a rectangular prism can be found using the formula $V = l \times w \times h$, where l is the length, w is the width and h is the height.
Vanessa is calculating the volume of a rectangular prism and converts all the dimensions to meters. What unit will the volume have?

15 The volume of a sphere can be found using the formula $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere.
Xavier is calculating the volume of a sphere with a radius measured in centimeters. What will the unit for the volume be?

16 The magnitude of the force F at work on an object with a mass of m kilograms and an acceleration of a meters per second squared can be found using the formula $F = ma$.
State the unit for force.

17 The density, D, of an object can be found using the formula $D = \frac{M}{V}$ where M is the mass of the object and V is the volume. Both mass and volume can use a variety of different units, as long as the same units are used when comparing the density of different objects.
If Rochelle used kilograms for the mass of an object and cubic centimeters for the volume of the object, what would the unit for density be in this case?

18 The time taken for an object to travel a certain distance can be calculated using the formula $t = \frac{d}{s}$, where s is the speed of the object and d is the distance traveled by the object.
If Roxanne measures speed in centimeters per minute and distance in centimeters, what will be the unit of time in her calculation?

19 Acceleration is a measure of how the velocity of a moving object is changing. It can be calculated using the formula $a = \frac{v - u}{t}$, where v represents the final velocity, u represents the initial velocity and t represents the time interval.
If the unit for the velocity of an object is kilometers per hour and time is measured in hours, what would the unit for the object's acceleration be?

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

Dimensional analysis

20 State the dimensions of the following in terms of length:

a Area **b** Volume **c** $\sqrt{\text{Area}}$

21 State the dimensions of acceleration in terms of length and time, if $\text{Acceleration} = \dfrac{\text{Velocity}}{\text{Time}}$ and $\text{Velocity} = \dfrac{\text{Length}}{\text{Time}}$.

22 State the dimensions of density in terms of mass and length if $\text{Density} = \dfrac{\text{Mass}}{\text{Volume}}$.

23 State the dimensions of force in terms of mass, length and time if $\text{Force} = \text{Mass} \times \text{Acceleration}$.

24 State the dimensions of pressure in terms of mass, length and time if $\text{Pressure} = \dfrac{\text{Force}}{\text{Area}}$.

25 State the dimensions of work in terms of mass, length and time if $\text{Work} = \text{Force} \times \text{Length}$.

26 State the dimensions of gravitational potential energy in terms of mass, length and time if $\text{Gravitational potential energy} = \text{Mass} \times \text{Acceleration} \times \text{Length}$.

27 For each of the following, determine whether F represents a length, an area or a volume:

a $F = As$ where A represents an area and s represents a length.

b $F = 3q + r$ where both q and r represent lengths.

c $F = p^2r$ where both p and r represent lengths.

d $F = 3\pi r^2$ where r represents a length.

e $F = 4\pi r^3$ where r represents a length.

f $F = \dfrac{2V}{A}$ where V represents a volume and A represents an area.

g $F = \dfrac{p^2r}{A}$ where both p and r represent lengths and A represents an area.

h $F = \dfrac{s^3}{p}$ where both s and p represent lengths.

i $F = \dfrac{A}{2\pi q}$ where A represents an area and q represents a length.

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

© ClipSAT: By Mr. Mohamed Abdallah +966597688647